

scalable conduit for idea diffusion, stimulating unexpected intellectual engagement and enriching scholarly discourse. Ironically, although the enormous volume of peer reviews conducted annually is often seen primarily as a burden, our results reveal that precisely because of this volume, peer review quietly yet substantially shapes the trajectory of scientific progress. This previously unrecognized mechanism underscores peer review's enduring and essential contribution—not merely as science's gatekeeper, but also as an influential force driving innovation and the advancement of diverse ideas.

Data Availability

The SciSciNet data are publicly available at <https://doi.org/10.6084/m9.figshare.c.6076908.v1>.

The Web of Science (WoS) data were used under license and are not publicly available. Peer review data contain sensitive information and cannot be shared publicly. The data necessary to reproduce the main figures in this study will be made available by the authors upon reasonable request.

Code Availability

The code used to generate the main figures and perform statistical analyses will be made available by the authors upon reasonable request.

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Author Contributions

B.W., D.W., and B.U. designed the research; B.W., and Z.M. collected data and conducted empirical analyses. All authors discussed the results and edited the manuscript.

Competing Interest Statement

The authors declare no competing interests.

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Supplementary Information for Peer Review and the Diffusion of Ideas

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